

Statement of Interest: KAIST Research Projects

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Project #1: I-driven Diagnostic Support System with Clinical Video Analysis

My work on the Context Isolation Framework (CIF) for Large Language Models has prepared me to address the critical challenge of developing reliable diagnostic support systems for tic disorders. Video-based diagnostic tools present a unique opportunity to revolutionize treatment through early detection and personalized intervention, but they require sophisticated contextual understanding to separate relevant from irrelevant movements.

Working with Dr. Zhenna Lu, I implemented attention mechanisms and semantic embeddings that could be directly applied to clinical video analysis. I propose adapting our hybrid isolation layers to separate involuntary tic movements from voluntary actions in video sequences. By leveraging techniques similar to those used in my pose detection project, we can create a system that:

1. Isolates contextually relevant movements through temporal attention mechanisms
2. Builds patient-specific movement profiles through embedding-based similarity analysis
3. Separates diagnostic signals from noise using principles derived from CIF

This approach aligns with recent findings from Pringsheim et al. (2024) on diagnostic criteria for tic disorders, which emphasize the importance of contextual factors in distinguishing tics from other movement disorders. By implementing a multi-modal framework that combines visual data with patient history embeddings, we can create a system that maintains contextual integrity across diagnostic sessions.

Project #2: Personalized AI Tutor: Adapting LLMs to Individual Learning Needs

The challenge of adapting LLMs to individual learning needs directly intersects with my work on context isolation and my experience leading the R&D Student Club. I've observed firsthand how contextual leakage in LLMs can lead to inappropriate generalization of learning patterns across different student profiles.

I propose extending my CIF framework to create learning context boundaries that:

1. Maintain separation between different learning styles without cross-contamination
2. Develop personalized knowledge graphs that evolve with student progress
3. Implement attention-based mechanisms for identifying knowledge gaps while preserving student-specific context

The RAG pipeline I developed for my Research Assistant project provides a foundation for retrieving contextually appropriate learning materials based on individual progress. By implementing isolation protocols between different student profiles, we can prevent the AI tutor from making false assumptions about knowledge transfer and learning pace.

Given my experience in both machine learning and context isolation protocols, I am uniquely positioned to contribute to either project. I am particularly excited about the opportunity to extend my work into visual AI and educational applications, areas that align perfectly with KAIST's research

excellence. I look forward to the possibility of bringing my Context Isolation Framework to these impactful applications under the guidance of KAIST's renowned faculty.